Where did the year go against plan — and what should planning do next?

A planning review of 52 weeks (23 Sep 2019 – 20 Sep 2020) across Online and Store channels, built on public H&M transaction data (Kaggle).

−13.3%

Sales vs plan. 418.0k actual against a 482.4k plan; −9.0% against last year.

35%

of the gap came from Stores, which held only 26% of last year's sales. Store sales fell to almost zero in April 2020.

−0.7%

Forecast bias, down from +10.8%, after adjusting last year's pattern for the recent 8-week trend. WAPE fell from 28.3% to 25.1%.

4.8 – 8.2

Weeks of cover if stock had been bought to plan: Store Accessories runs short, Online Shoes sits heavy.

RECOMMENDATIONS

01 **Re-base the next plan on trend-adjusted demand**, not last year +5%. The fixed uplift alone explains 23.0k (36%) of the miss.

02 **Rebalance receipts by channel**: slow Online Upper/Lower body, protect Store Accessories, Full body and Underwear, where late-summer demand ran 7–25% above plan.

03 **Plan Stores with wider ranges**: keep the seasonal model as a benchmark and review Store exceptions weekly, where error stays at 34%.

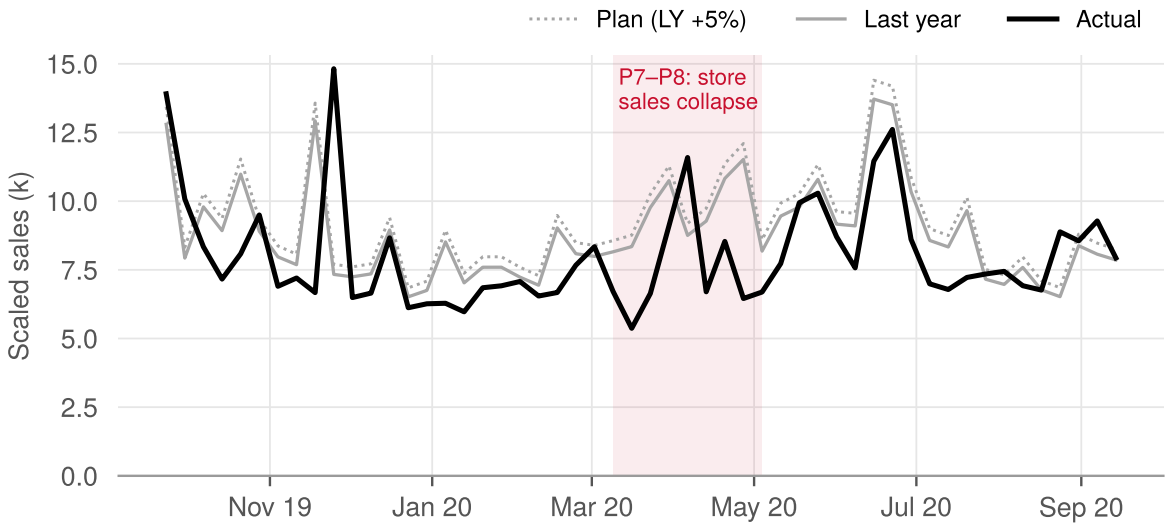
CONTENTS

Sales performance	01
Plan variance	02
Forecast accuracy	03
Inventory & receipts	04
Method, data & limits	05

Sales finished 13.3% below plan: Online held close to last year, Stores did not

ACTUAL SALES	VS PLAN	ONLINE VS LY	STORE VS LY
418.0k	−13.3%	−5.5%	−19.2%
Scaled value, 52 weeks	−64.3k against LY +5% target	Share 77.1% (+2.9 pts)	Near zero in Apr 2020, then above LY by Aug

WEEKLY ACTUAL VS LAST YEAR VS PLAN · BOTH CHANNELS



What happened

Sales tracked below last year from October. The steepest drop came in March–April 2020, when Store sales fell by 80–100% across two periods; Online softened less. Stores then recovered to 17–18% above last year in August–September, while Online stayed below.

CATEGORIES TOP 7 BY SALES

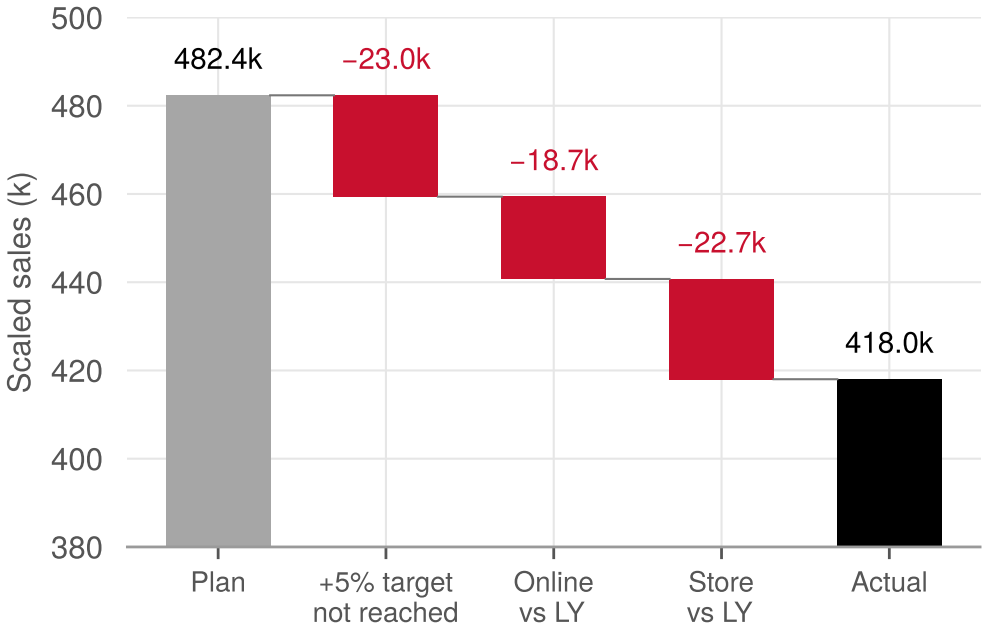
CATEGORY	SALES	MIX	VS LY	VS PLAN	ONLINE
Upper body	158.1k	38%	−10.8%	−28.1k	73%
Lower body	109.0k	26%	−9.8%	−17.8k	79%
Full body	62.9k	15%	−3.1%	−5.2k	86%
Underwear	28.1k	7%	+8.1%	0.8k	72%
Swimwear	25.4k	6%	−20.8%	−8.3k	87%
Shoes	12.4k	3%	−23.5%	−4.6k	90%
Accessories	11.5k	3%	−12.6%	−2.3k	59%

Focus

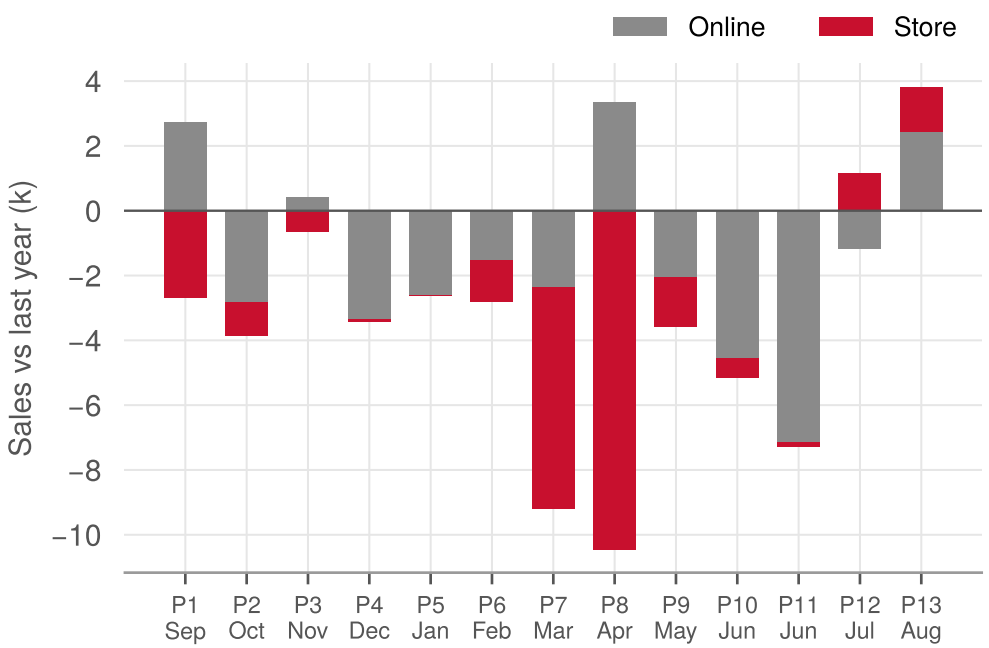
Upper and Lower body are 64% of sales and 71% of the gap to plan. Underwear was the only large category to grow. Swimwear and Shoes fell over 20%, the steepest drops in the top seven.

About a third of the miss was the target itself; the rest was demand, concentrated in Stores in spring

BRIDGE PLAN TO ACTUAL, SCALED SALES



TIMING SALES VS LAST YEAR BY 4-WEEK PERIOD



Target (−23.0k)

The plan assumed +5% on last year in every week and series. Even a flat year would have missed by this amount, so this part is a target-setting question, not a trading one.

Online (−18.7k, −5.5%)

A steady shortfall spread across most periods rather than one event. This points to a lower demand base, which the trend-adjusted forecast (p.03) picks up.

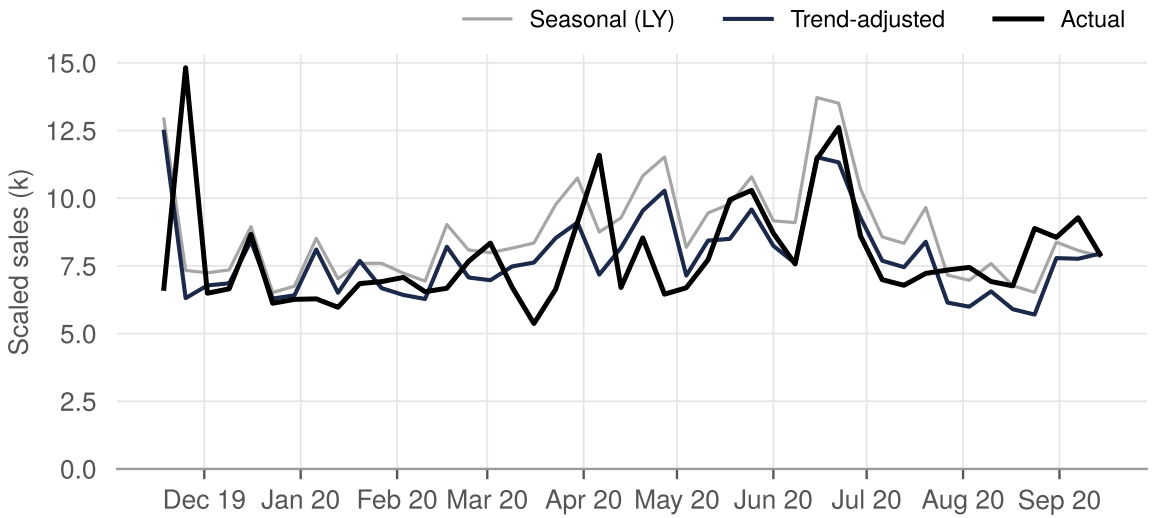
Store (−22.7k, −19.2%)

Mostly P7–P8 (9 Mar – 3 May 2020), when Store sales fell 80% then 100%. This coincides with the COVID-19 period; the data alone cannot confirm the cause.

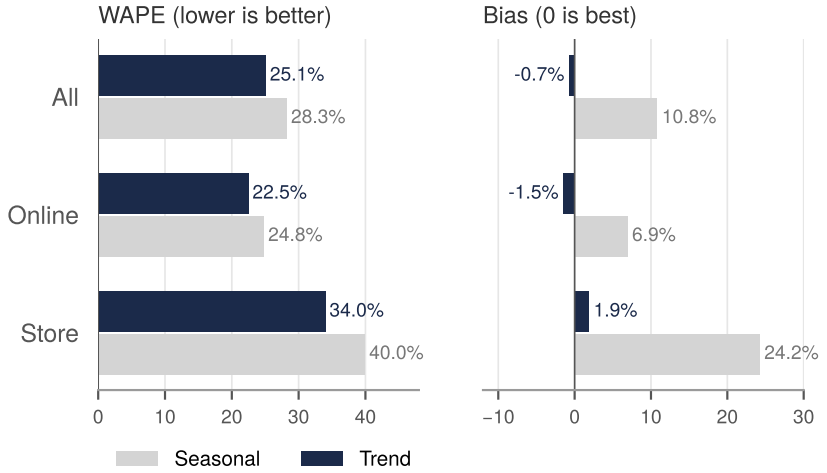
Adjusting last year for the recent trend removed an 11% overforecast

WAPE, SEASONAL	WAPE, TREND-ADJUSTED	BIAS, TREND-ADJUSTED	STORE WAPE
28.3%	25.1%	−0.7%	34.0%
Same week last year	−3.2 pts	from +10.8%	Still high; Online 22.5%

WEEKLY ACTUAL VS THE TWO FORECASTS · BOTH CHANNELS



BY CHANNEL SALES ERROR



Method

Trend forecast = last year's week × (last 8 weeks ÷ same 8 weeks a year earlier), capped at 0.5–1.5. Only data available before the week is used.

What it means for planning

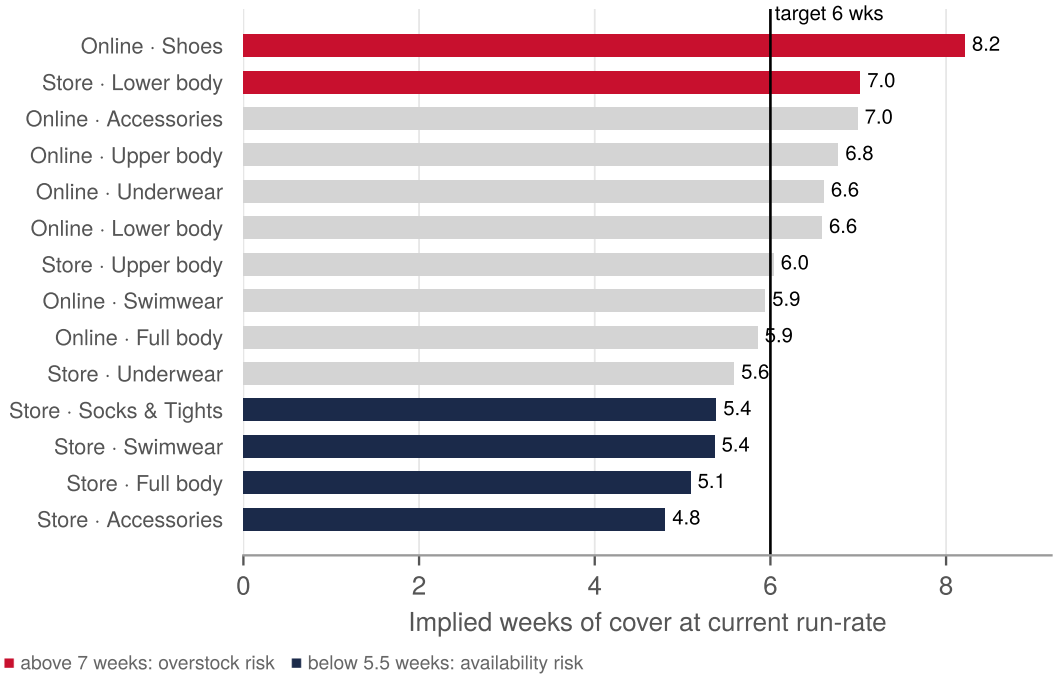
The seasonal model kept planning for last year's demand. The trend factor corrects the level within a few weeks, which matters most for receipts that are not yet committed.

Where it falls short

It lags sudden shocks like the store drop in spring, then over-corrects on recovery. For Stores, keep the seasonal benchmark and review large exceptions manually.

A plan-based buy would leave Online heavy and Stores short going into autumn

WEEKS OF COVER IF STOCK WAS BOUGHT TO 6 WEEKS OF PLAN DEMAND



RECEIPT ACTION UNITS TO BRING EACH SERIES BACK TO 6 WEEKS

SERIES	DEMAND VS PLAN	COVER	RECEIPT CHANGE
Store · Accessories	+25%	4.8	+10k units
Store · Full body	+18%	5.1	+7k units
Store · Swimwear	+12%	5.4	+2k units
Store · Socks & Tights	+11%	5.4	+2k units
…			
Online · Upper body	−11%	6.8	−54k units
Online · Accessories	−14%	7.0	−5k units
Store · Lower body	−15%	7.0	−17k units
Online · Shoes	−27%	8.2	−9k units

Why this matters

Stores came back above plan in late summer while Online stayed below. Buying to last year +5% would tie up stock in Online Upper and Lower body and leave Store basics short. A channel-level rebalance moves stock to where demand is now, before buying more in total.

Assumption

No inventory data exists in this dataset. Opening stock is assumed to equal 6 weeks of plan demand (LY units +5%); cover = that stock ÷ actual weekly units. Validate against real stock, lead times and minimum orders.

How this was built

DATA

H&M Personalized Fashion Recommendations (Kaggle): 31.8M transaction rows, aggregated to 104 complete Monday-start weeks × 2 channels × 19 product groups (36 active series). Duplicate keys: 0. Weekly grid validated.

Prices are a scaled index, not EUR. One transaction row = one unit. Missing combinations are filled as zero sales.

DEFINITIONS

- **Plan**: same week last year × 1.05, fixed before the year starts.
- **Seasonal forecast**: same week last year.
- **Trend forecast**: seasonal × capped 8-week YoY ratio.
- **WAPE / Bias**: errors summed at series × week level, then divided by actual sales.
- **Periods**: 13 × 4 weeks, P1 starts 23 Sep 2019.

LIMITS

- No country, stock, cost, markdown or margin data; page 04 is a scenario.
- Forecast parameters (8 weeks, 0.5–1.5 cap) are fixed, not tuned.
- Causes of sales changes are not established from the data.
- Daily completeness of the source is not verified.

PIPELINE

```
transactions (31.8M rows)
→ weekly_sales.csv       aggregate & validate
→ build_forecast.py      plan, forecasts, accuracy
→ planning_weekly.csv    3,744 rows → Power BI model
→ this review
```

TOOLS

Python (pandas, numpy) for data preparation and backtesting · [Power BI: dashboard with channel/category slicers] · [Excel: weekly plan & receipt worksheet]